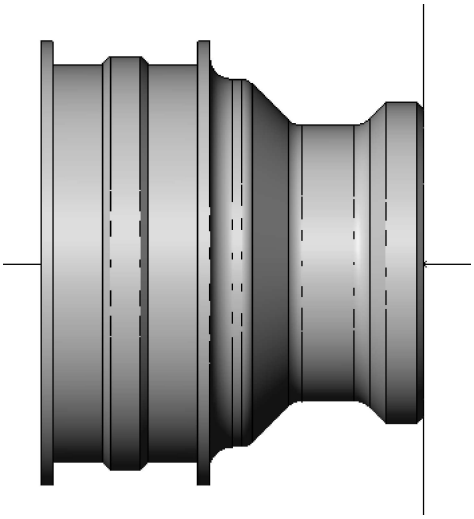
Import a turning part and get it ready for machining

Before you can start creating turning operations, you need to define your part and stock. ESPRIT maintains awareness of the stock model as turning operations are applied. As stock is removed, the stock model is updated and new turning operations generate toolpath only in areas where stock remains.

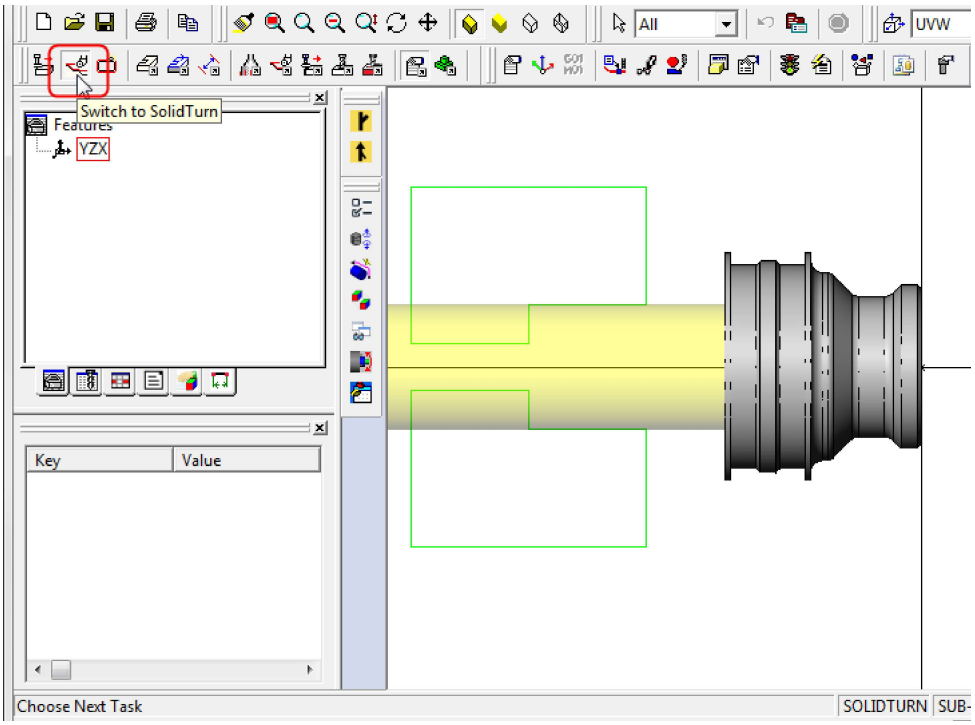
1. Start a new session of ESPRIT.
2. On the **Tools** menu, make sure **System Unit** is set to **Metric**.
3. Click **File** > **Open**.
4. Navigate to the folder **05-Turning** and open the file **TurnedPart.x_t**.



5. On the Status bar, make sure **HI** mode and **SUB-ELEMENTS** mode are enabled.

For this part, you will define the stock as a cylindrical bar. Bar stock for lathes is defined in SolidTurn Machine Setup.

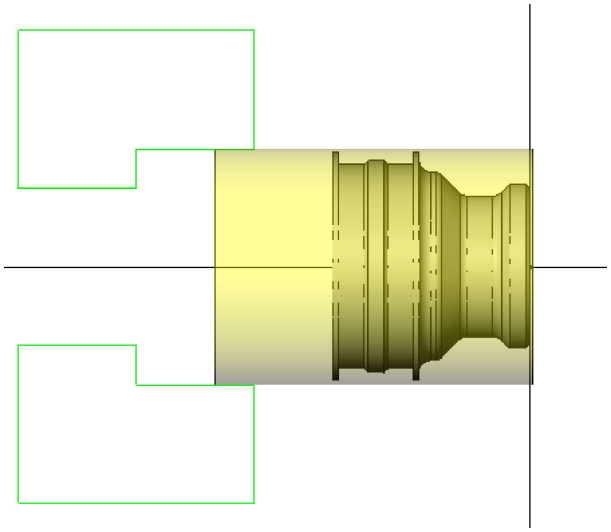
1. On the Smart toolbar, click **Switch to SolidTurn**. Notice that lathe stock is now displayed in the graphics area. Also notice that the work coordinate in the Feature Manager changes to YZX. Because ESPRIT supports both milling and turning, the X axis is used as the turning axis in ESPRIT.



2. Click **Common Machining** and then click **Machine Setup**.

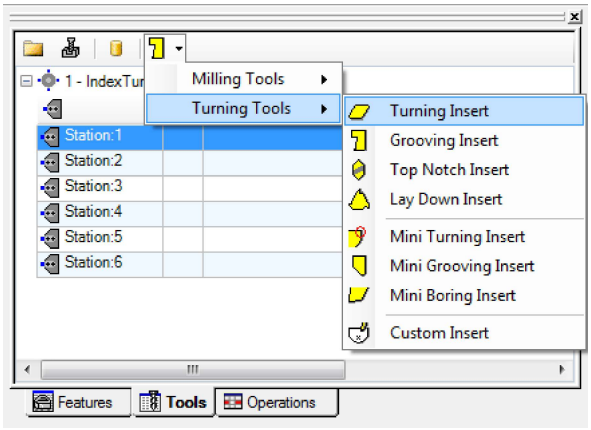


3. Change **Bar Diameter** to **60**.
4. Change **Total Bar Length** to **81**.
5. Change **Part Stock Length** to **50**.
6. Change **Start Position Z** to **1**. This moves the start position of the bar forward to create 1 mm stock allowance on the face of the part.
7. Click **OK** to update the stock and close Machine Setup.



Mount tools on the turning machine

Commands in the Tool Manager let you create milling and turning tools.



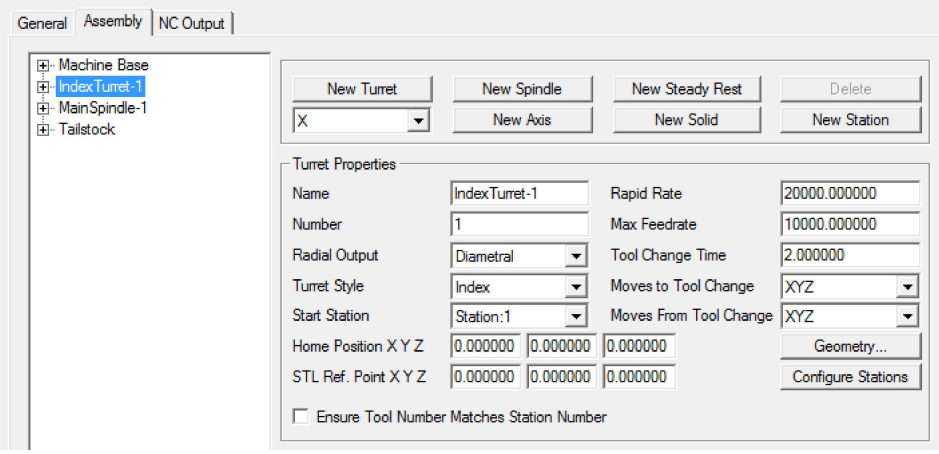
You choose the type of tool you want to create, choose the turret and station where the tool will be mounted on the machine, and define dimensions for the insert and shank.

You will start by modifying the standard machine setup in ESPRIT to change the configuration of the turret and add more tool stations. The default turret in SolidTurn Machine Setup is an index turret with six tool stations. You will add another six stations for a total of twelve.

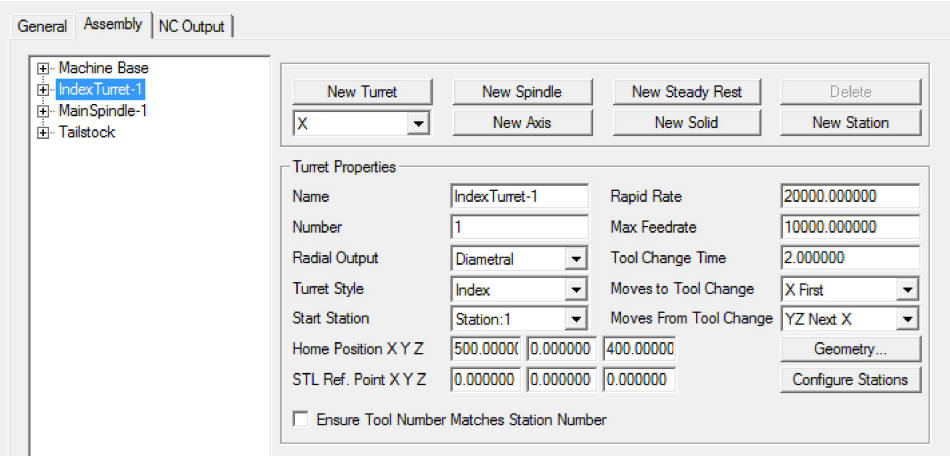
1. Click **Common Machining** and then click **Machine Setup**.



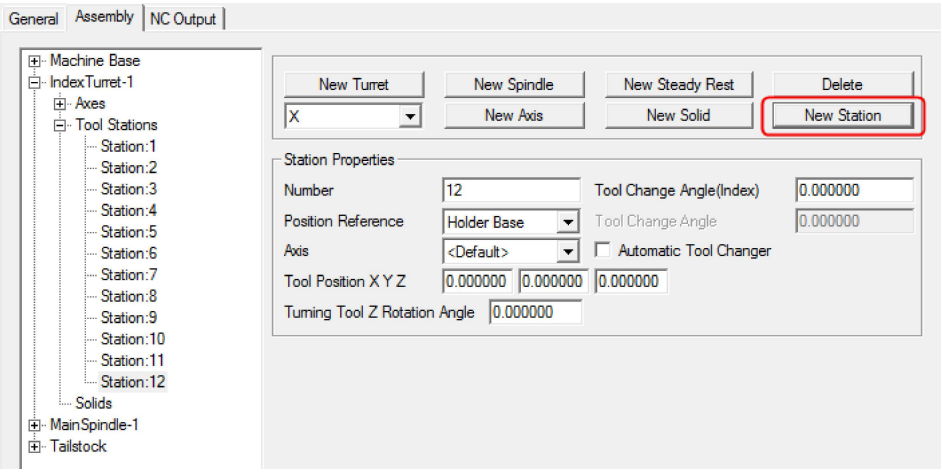
2. Click the **Assembly** tab and select **IndexTurret-1** to view its properties.



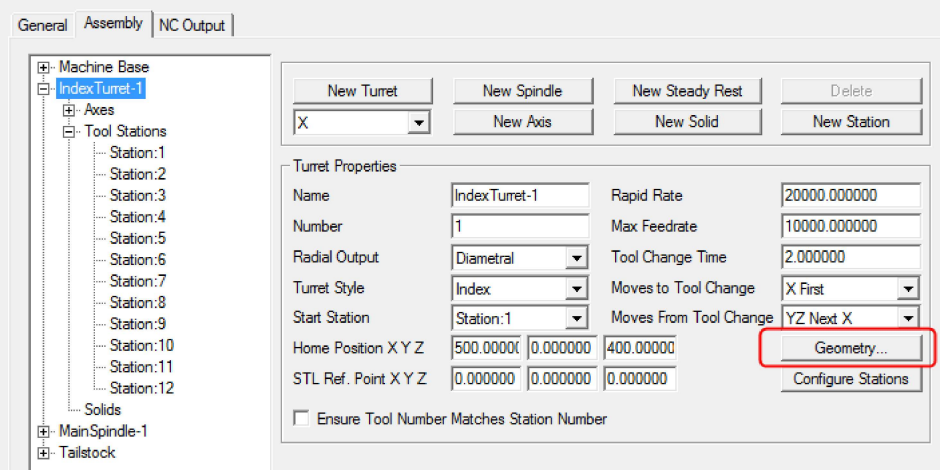
3. Change **Moves to Tool Change** to **X First**.
4. Verify that **Moves From Tool Change** has changed to **YZ Next X**.
5. Change **Home Position X Y Z** to **500**, **0**, **400**.



6. Expand IndexTurret-1 in the tree to view the number of tool stations.
7. Click the **New Station** button until there are a total of 12 tool stations on the turret. Notice that the tool change angle and tool position for each new station is set to zero. Instead of entering properties for each tool station manually, you will learn how to configure the tool stations automatically.



8. Select **IndexTurret-1** in the tree and then click the **Geometry** button.

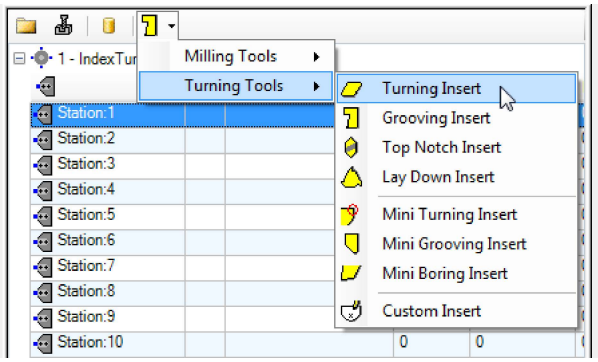


9. In the dialog, change **Turret Diameter** to **420** and **Turret Width** to **0** and click **OK**.
10. Click the **Configure Stations** button to configure the new tool stations on the updated turret geometry.
11. Now select Station:12 in the tree and notice that the tool change angle and tool position have been set.
12. Click **OK** to save your changes and close Machine Setup.

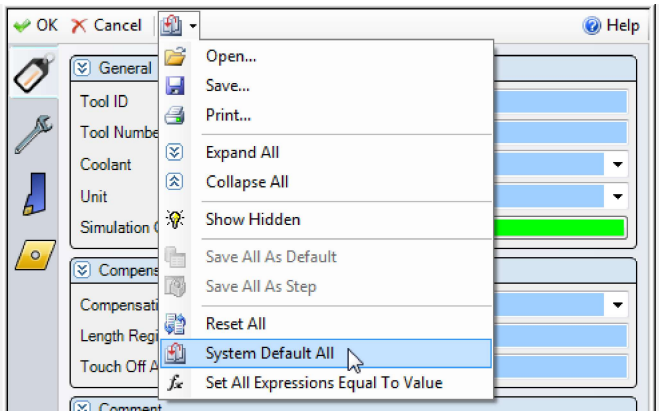
Create turning tools

The first tool you create will be used for roughing on the face of the part. It will be mounted in a vertical orientation on the turret, facing the main spindle.

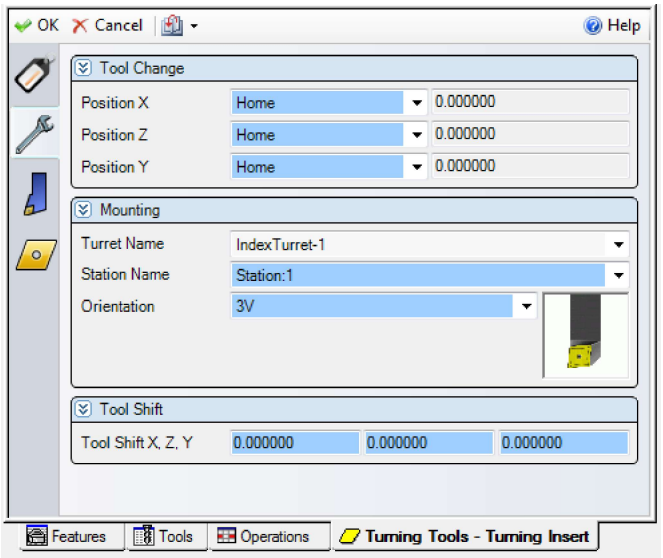
1. In the Tools Manager, click **Turning Tools** and then select **Turning Insert**.



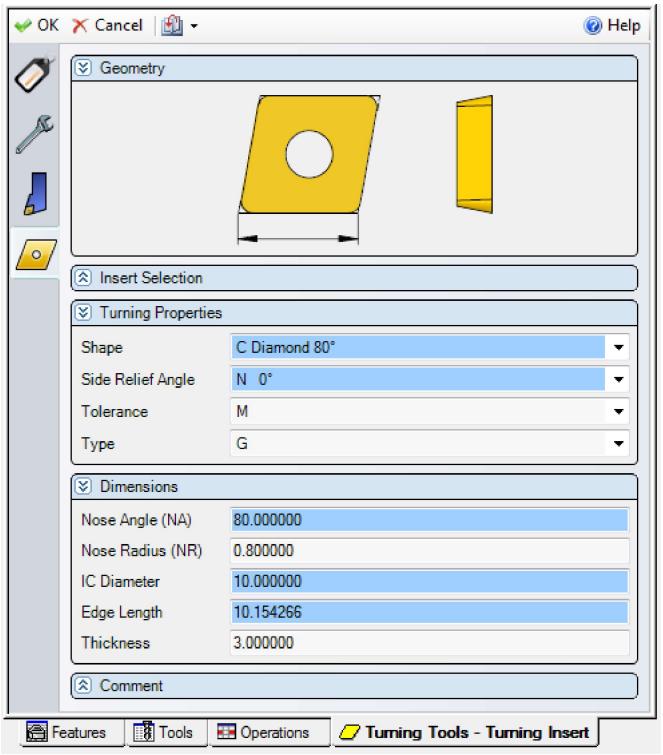
2. From the dropdown menu, click **System Default All** to reset all parameters to the system defaults.



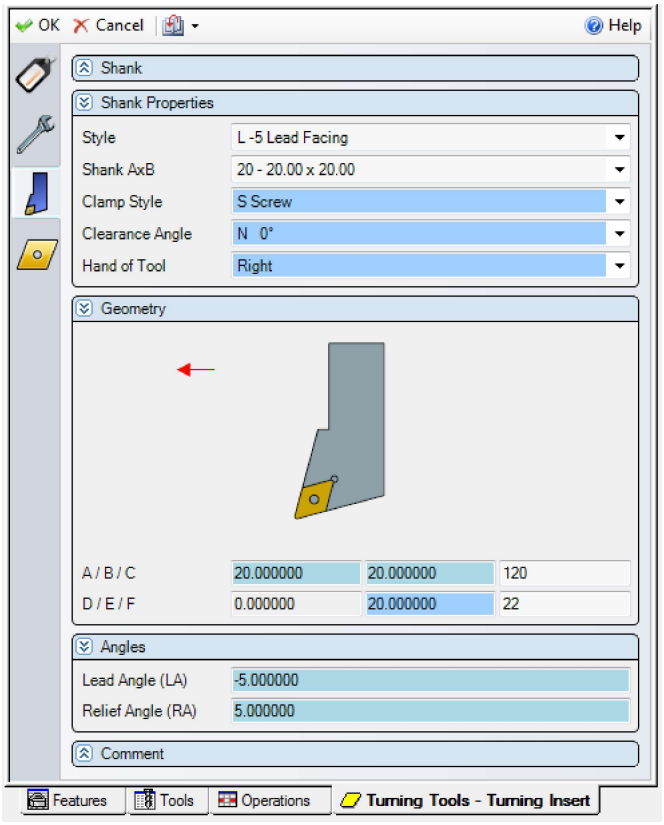
3. Click the **General** tab and change the following parameters:
   - Tool ID = **OD Rough**
   - Simulation Cut Color = **Yellow**
   - Click the **Settings** tab and change the following parameters:
   - Turret Name = **IndexTurret-1**
   - Station Name = **Station:1**



4. Click the **Insert** tab and change the following parameters:
   - Tolerance = **M**
   - Type = **G**
   - Nose Radius = **0.8**
   - Thickness = **3**



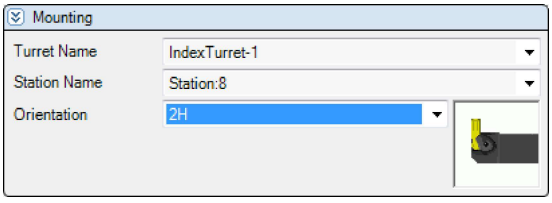
5. Click the **Shank** tab and change the following parameters:
   - Style = **L-5 Lead Facing**
   - Shank AxB = **20 - 20.00 x 20.00**
   - C = **120**
   - F = **22**



6. Click **OK**.

This next tool is a full radius grooving insert that will be used on the inner diameter of the part. It will have a round shank and be mounted in a horizontal orientation on the turret.

1. Click **Turning Tools** and select **Grooving Insert**.
2. Click **System Default All**.
3. On the **General** tab:
   - Tool ID = **ID Groove R4**
   - On the **Settings** tab:
   - Turret Name = **IndexTurret-1**
   - Station Name = **Station:8**
   - Orientation = **2H**



4. On the **Insert** tab:
   - Shape = **Full Radius**
   - Nose Angle = **90**
   - Nose Radius = **4**
   - Size = **10**
   - E = **8.5**
   - Thickness = **2**
5. On the **Shank** tab:
   - Shank Type = **Round Shank**
   - Tool Diameter = **10 - 10.00**
   - A = **16**
   - C = **55**
   - F = **13.5**
6. Click **OK**.

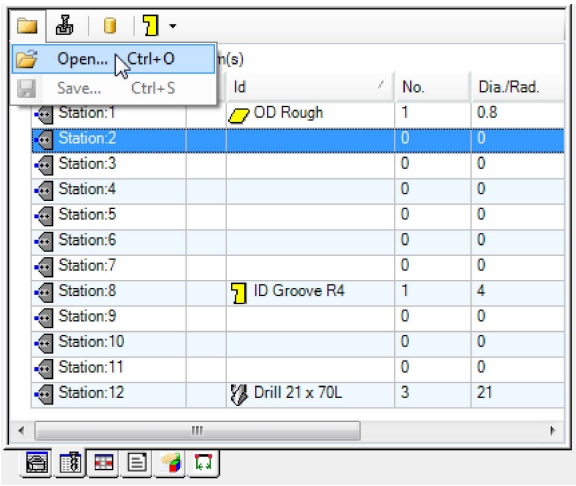
Create a twist drill

1. Click **Milling Tools** > **Drill**.
2. Click **System Default All**.
3. On the **General** tab:
   - Tool ID = **Drill 21 x 70L**
   - Simulation Cut Color = **Orange**
4. On the **Settings** tab:
   - Turret Name = **IndexTurret-1**
   - Station Name = **Station:12**
   - Axis Orientation = **Z+**
5. On the **Shank** tab:
   - Shank Diameter = **21**
   - Tool Length = **70**
   - Cutting Length = **60**
6. On the **Cutter** tab:
   - Tool Diameter = **21**
7. On the **Extension** tab:
   - Overall Length = **70**
8. Click **OK**.

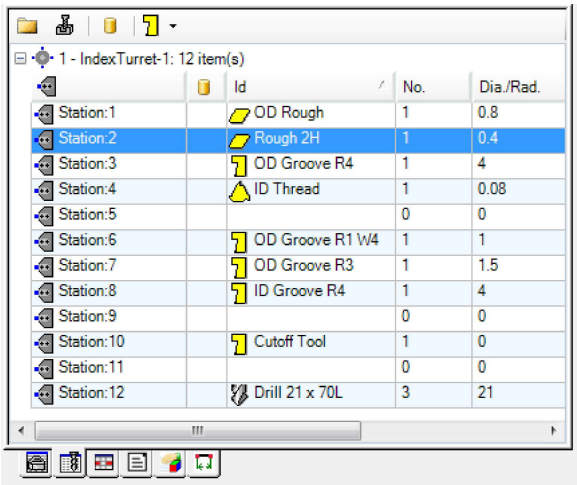
Import tools from an external file

Any cutting tools in the Tool Manager can be saved to an ESPRIT Tool Library (*.etl) file. One or more tools can be selected and saved to a file. The tools in the file can then be imported into any other ESPRIT document.

1. In the Tool Manager, click **File** > **Open**.



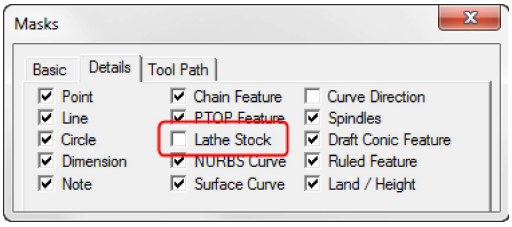
2. Open the file **TurningTools.etl**, located in your lesson folder. The tools in the file are added to the tool list in the Tool Manager.



Create features for turning

You will use the Turning Feature Recognition command on the Accessories toolbar to analyze the shape of the solid model to find internal and external profiles. Turning Feature Recognition rotates the part around the turning axis (the X-axis in ESPRIT) to find the true turned profile.

1. Press **Ctrl+M** to display the Masks dialog, click the **Details** tab and then select **Lathe Stock** to mask the display of the stock. Close the dialog.

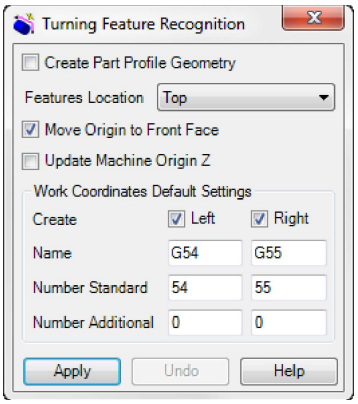


2. Make sure the **Accessories** toolbar is visible. If the toolbar is not displayed, click **Tools** > **Add-In** to display the dialog. Choose **Accessories Add-In**, select **Loaded** and **Load on Startup**, and then click **OK**.

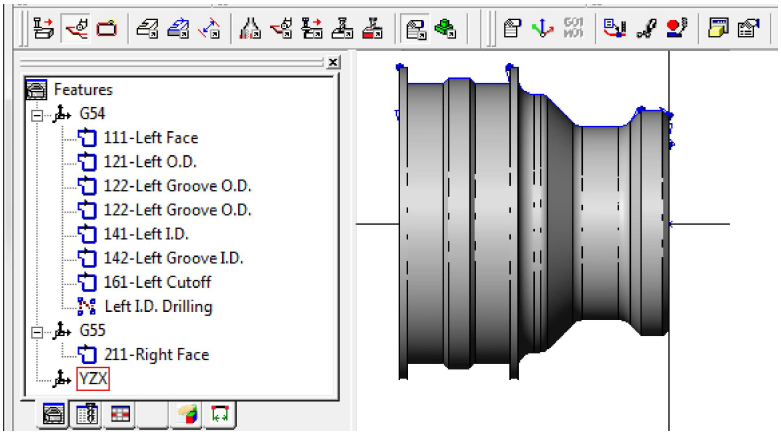
3. On the **Accessories** toolbar, click **Turning Features**.



4. Use the default settings in the dialog as shown below.



5. Select the solid model and click **Apply**. Turning features are recognized and assigned an appropriate name and work coordinate.



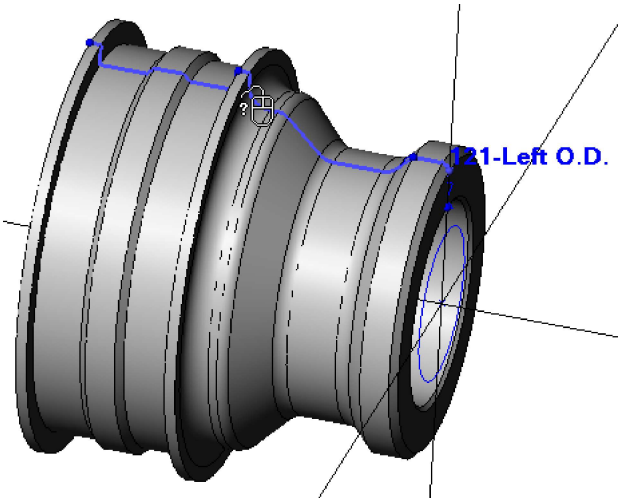
6. Close the dialog.

Chain features can be modified to add new elements, delete elements, reverse the direction, and change the start point. You will learn how to modify the chain on the OD profile to replace the elements in the groove with a straight segment over the top of the groove.

1. On the Smart toolbar, click **Features** and then click **Manual Chain**. The Manual Chain command can be used to modify chain features as well as create them.



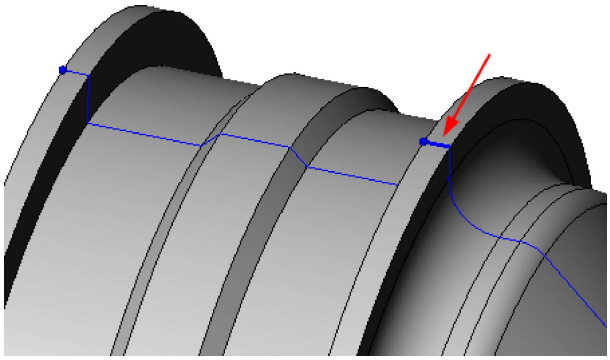
2. Select the feature on the OD. The chain can now be edited to add and remove elements. The cursor is attached to the end point of the chain.



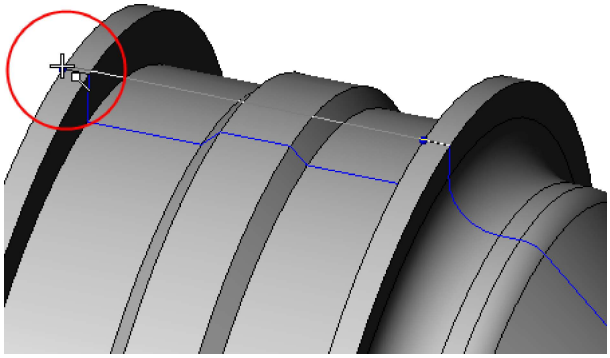
3. On the **Edit Feature** toolbar, click **Move Back**.



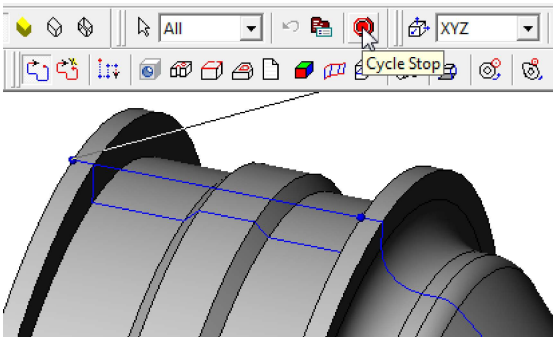
4. Select the feature sub-element shown to remove all elements between that element and the end point of the chain.



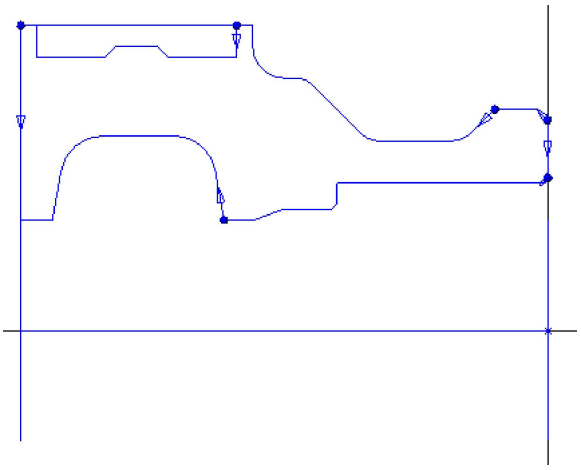
5. Now select the end point shown to extend the feature over the groove.



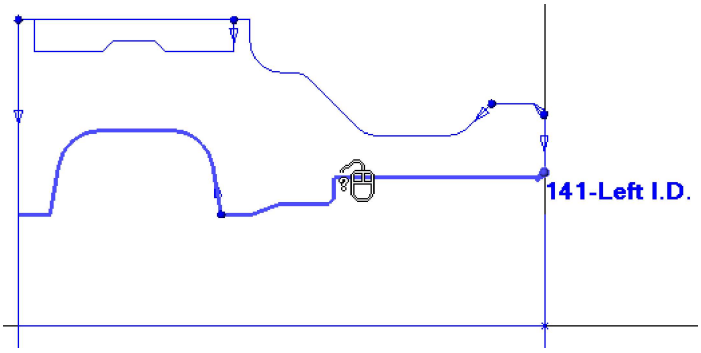
6. Click **Cycle Stop** to confirm the change.



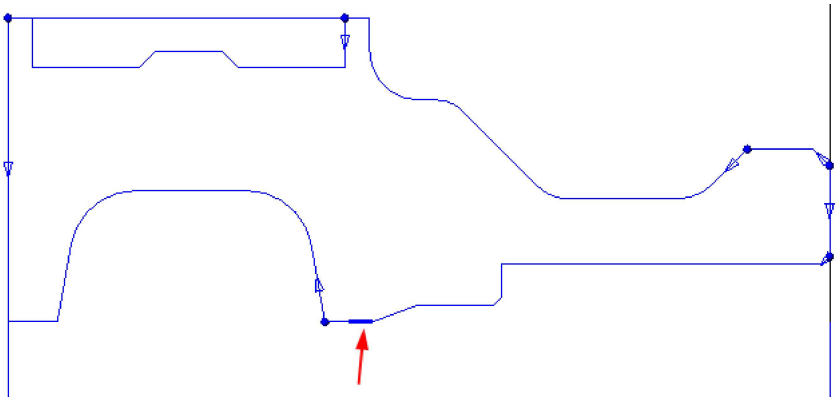
7. Mask the solid model.
8. Change the view to **Top** and zoom in to the profiles.



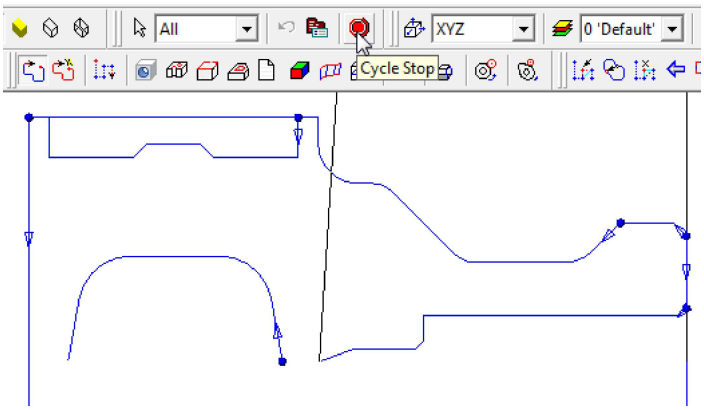
9. Click **Manual Chain** again and select the chain on the ID profile.



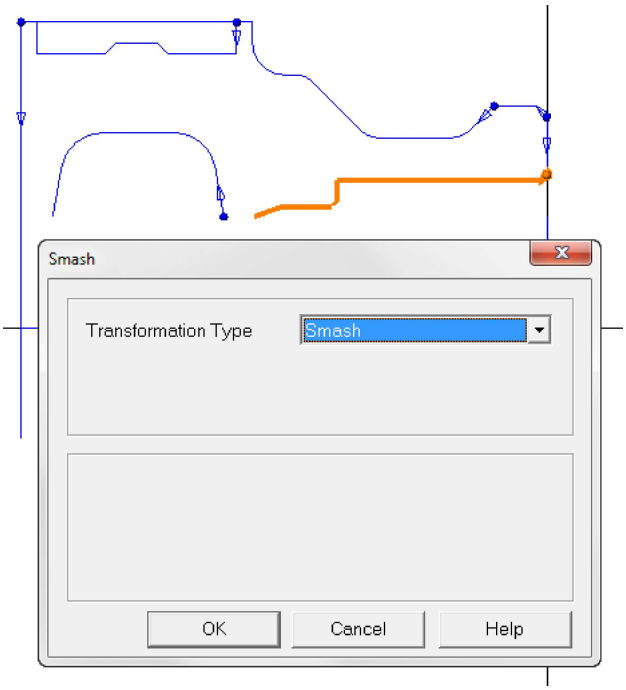
10. Click **Move Back** and use Highlight mode to select the feature sub-element shown.



11. Click **Cycle Stop** to confirm the change.

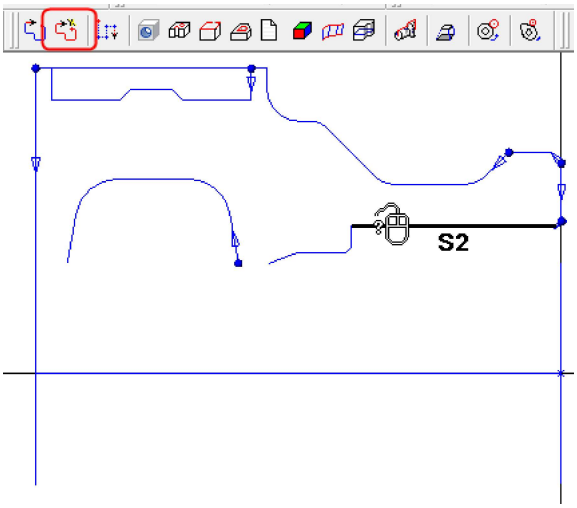


12. Select the ID feature and press **Ctrl+C**. Set **Transformation Type** to **Smash** and click **OK** to extract wireframe geometry from the chain. You will use a segment to create a simple chain feature for a threading operation at the front of the ID.

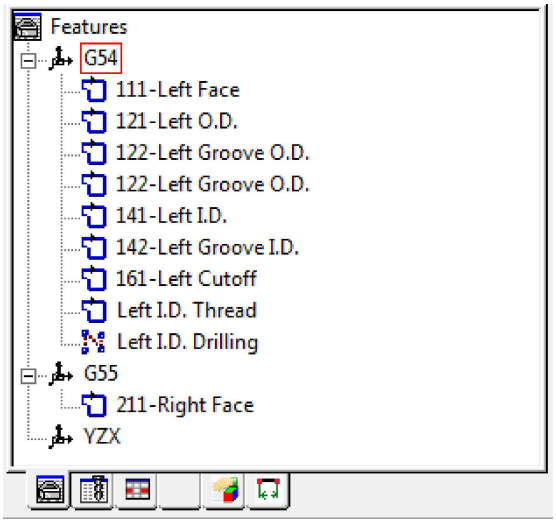


13. In the Feature Manager, double-click on **G54** to activate the work coordinate. When you create a new feature, it will be created on this work coordinate.

14. Use Highlight mode to select just the segment at the front of the ID and click **Auto Chain**.



15. In the Feature Manager, right-click on the new chain feature and then click **Rename**. Rename the feature **Left I.D. Thread**.



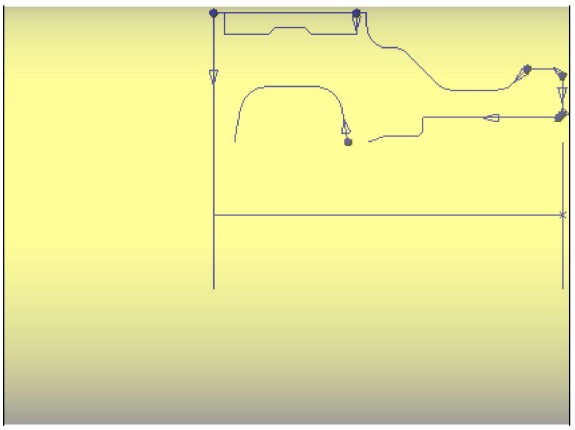
Rough the face

You will use a roughing operation to quickly remove stock material on the face of the part. This is accomplished with linear cutting passes at incremental depths.

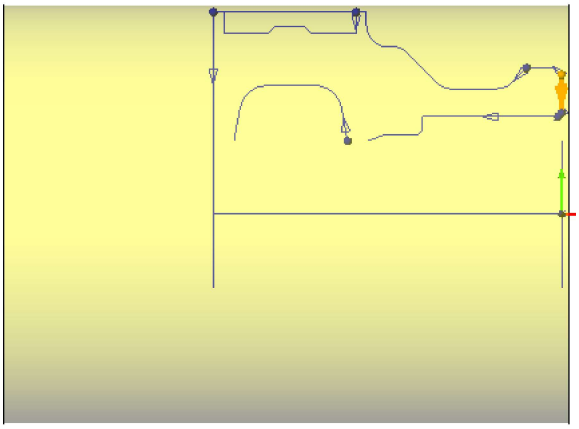
Roughing uses stock automation to identify areas where material needs to be removed. The system starts with the initial stock definition in Machine Setup and maintains awareness of all material removed by turning operations up to the current operation.

The feature on the face of the part is extremely short. However, this is not a problem because turning operations let you extend the toolpath to start cutting outside the bar stock and end at a point below the turning axis.

1. Press **Ctrl+M** and unmask **Lathe Stock.**



2. In the Feature Manager, select the feature **Left Face**.

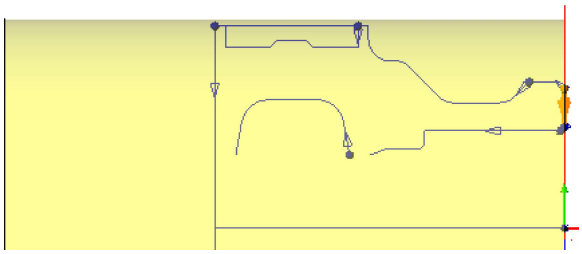


3. On the Smart toolbar, click **Solidturn** and then click **Roughing**.



4. Click **System Default All** to reset all parameters to the system defaults.
5. On the **General** tab:
   - Operation Name = **Face Rough**
   - Tool = **OD Rough**
   - Speed CSS = **360** (speed and feed values are required)
   - Feedrate PR = **0.1**

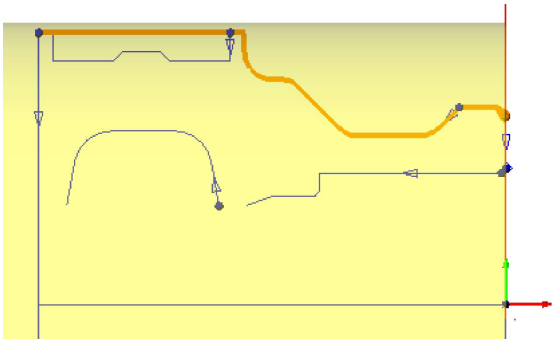
6. On the **Strategy** tab:
   - Type of Work = **Face**
   - End Extension = **16.5** (this value is measured from the end point of the feature to extend the toolpath below the centerline)
7. On the **Rough** tab:
   - Stock Type = **Automation** (the system calculates the lathe stock automatically, starting from the initial bar stock)
   - Rough Stock Z, X = **0**, **0**
   - Maximum Depth of Cut = **2**
   - Lead-In Type = **Tangent** (since this is a face cut, it is better to lead in with a tangent move)
8. Click **OK**. Notice that the lathe stock is updated.



Rough the OD

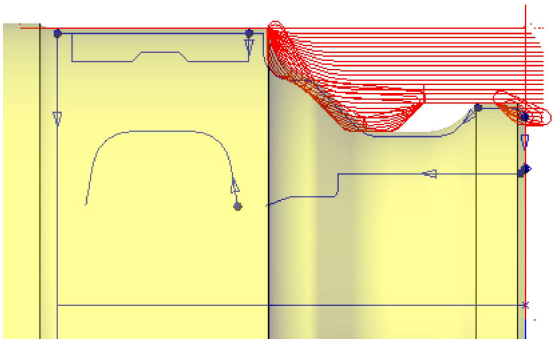
When you rough the profile on the outer diameter, you will use a ProfitTurning strategy. ProfitTurning is a high-speed roughing strategy that significantly reduces cycle times while increasing tool life. The ProfitTurning strategy blends trochoidal cutting passes with traditional linear passes. Arcs and trochoidal tool motion are inserted at sharp corners to maintain a constant tool load, which is ideal for hard material machining.

1. Select the feature **Left OD**.



2. Click **Roughing**. You will use many of the parameters from the face roughing operation to rough the OD. ESPRIT retains the settings from the last roughing operation until you reset them.
3. On the **General** tab:
   - Operation Name = **OD Rough**
   - Tool = **OD Groove R4** (The ProfitTurning strategy only supports circular inserts, such as this full radius groove tool)
   - Speed CSS = **800**
   - Feedrate PR = **.032**

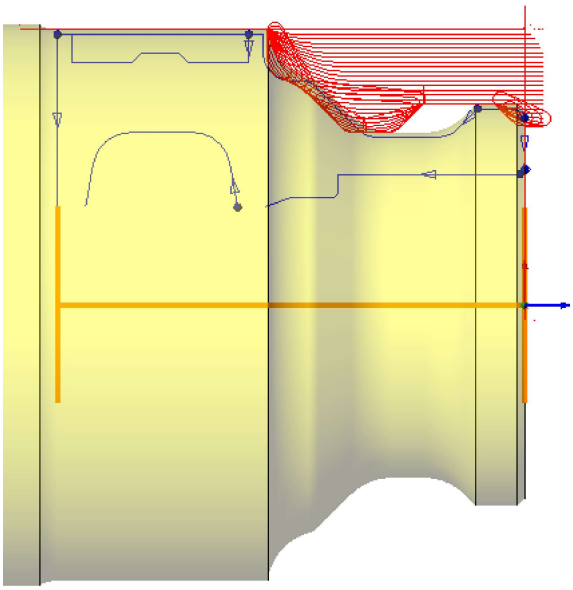
4. On the **Strategy** tab:
   - Type of Work = **OD**
   - End Extension = **0**
5. On the **Rough** tab:
   - Rough Stock Z, X = **0.5**, **0.5**
   - Rough Pattern = **ProfitTurning - Parallel**
   - Maximum Depth of Cut = **.5**
   - Min Trochoidal Radius, % = **2.0** (25%)
   - Alternate Cut Direction = **No** (ProfitTurning supports zigzag motion in the toolpath)
   - Lead-In Type = **Normal**
   - Normal Distance = **0.2**
   - Lead-Out Type = **None**
6. Click **OK.** Notice the circular motion in the toolpath where the tool contacts corners and walls.



Drill on the center line

To prepare for turning operations on the ID, you will first drill a hole that has the same diameter as the smallest diameter on the ID. Lathes are capable of drilling on the center line of the part. Lathes equipped with a Y axis are capable of off-center drilling. For this lesson, you will simply drill on the center line.

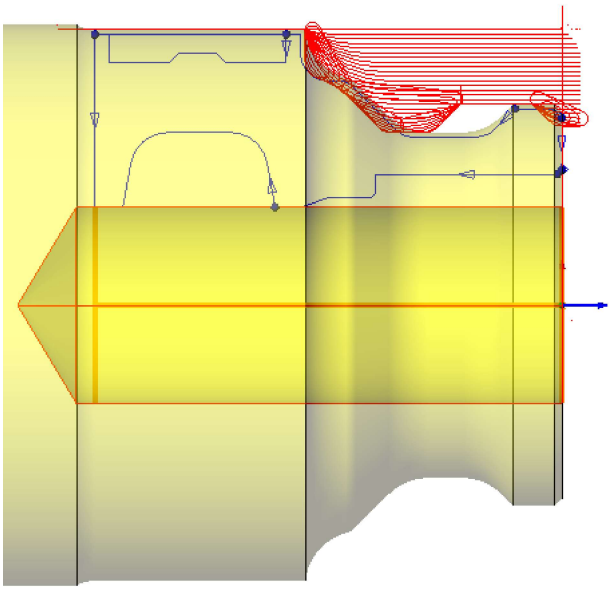
1. In the **Feature Manager**, select the feature **Left I.D. Drilling**.



2. Click **Drilling**.



3. Click **System Default All**.
4. On the **General** tab:
   - Operation Name = **Center Drill**
   - Tool = **Drill 21 x 70L**
   - Speed RPM = **2000**
   - Feed Unit = **Per Revolution**
   - Feedrate PR = **0.1**
5. On the Drill tab, notice that Total Depth is automatically set to the depth of the feature.
6. Click **OK**.

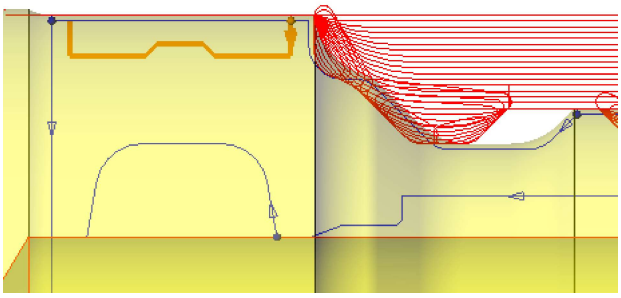


Rough and finish the groove on the OD

The Grooving cycle lets you create rough and finish passes in the same operation. Rough passes in the Grooving cycle also support the ProfitTurning strategy. In this lesson, the groove is roughed with multiple plunge moves at 80% of the width of the tool.

The finish pass is a contouring pass that follows the feature. In this case, the finish pass is split into two passes that start on each side of the groove and meet in the middle.

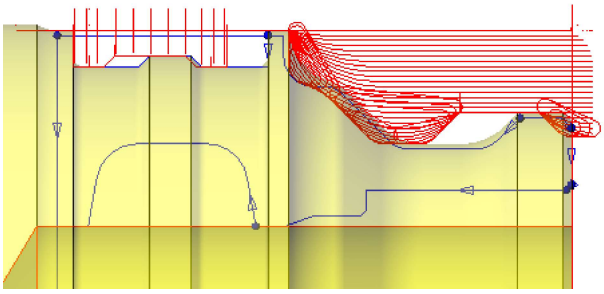
1. Select the feature **Left Groove O.D.**.



2. Click **Grooving**.



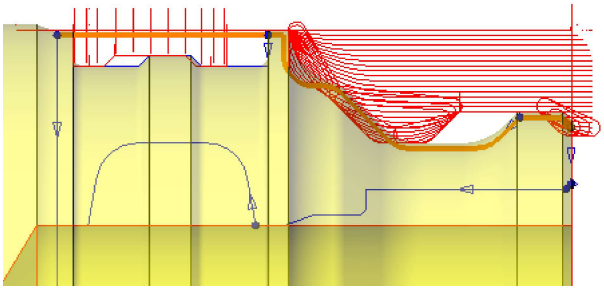
3. Click **System Default All**.
4. On the **General** tab:
   - Operation Name = **OD Groove**
   - Tool = **OD Groove R1 W4**
   - Speed CSS = **800**
   - Feedrate PR = **0.032**
5. On the Strategy tab, notice that Rough Pass and Finish Pass are both set to Yes.
6. On the **Rough** tab:
   - Stock Type = **Automation**
   - Rough Stock = **0.2**, **0.2**
   - Groove Type = **Multiple Plunge**
   - Step Over = **1.8** (Press the Tab key to calculate the percent of the tool diameter)
7. On the **Finish** tab:
   - Finish CSS = **200**
   - Finish PM = **0.1**
   - Lead-In Type = **Z and X Offset**
   - Offset Z, X = **0**, **1.0**
   - Lead-Out Type = **Z and X Offset**
   - Offset Z, X = **0**, **1.0**
8. Click **OK**.



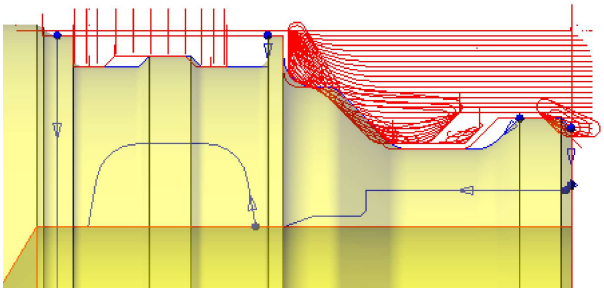
Re-rough and finish the OD

Roughing operations also let you create a single operation that includes roughing passes as well as a finish contouring pass. The same tool is used for entire operation.

1. Select the feature **Left OD**.



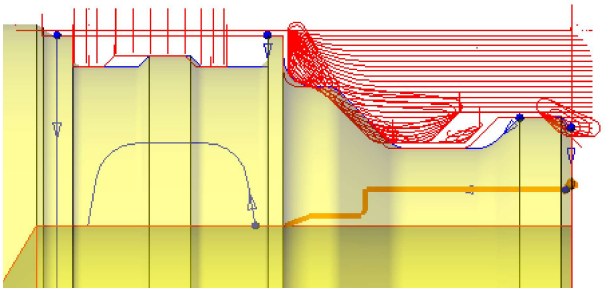
2. Click **Roughing**.
3. On the **General** tab:
   - Operation Name = **OD Re-rough and Finish**
   - Tool = **OD Groove R3**
4. On the **Strategy** tab:
   - Finish Pass = **Yes** (notice that the Finish tab displays)
5. On the **Rough** tab:
   - Alternate Cut Direction = **Yes**
   - Lead-In Type = **Z and X Offset**
   - Offset Z, X = **0**, **2**
   - Lead-Out Type = **Z and X Offset**
   - Offset Z, X = **0**, **2**
6. On the **Finish** tab:
   - Finish CSS = **800**
   - Finish PR = **0.032**
   - Lead-In Type = **Tangent**
   - Tangent Distance = **2.0**
   - Lead-Out Type = **Normal**
   - Normal Distance = **2.0**
7. Click **OK**.



Rough and finish the ID

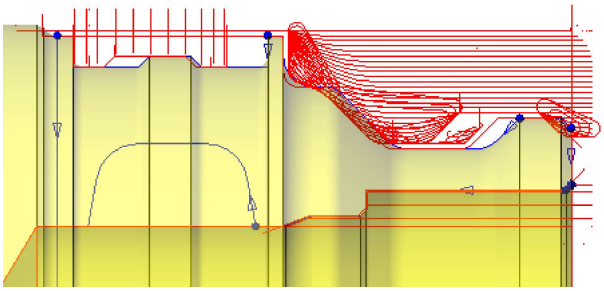
You will use a roughing operation with standard incremental passes to rough and then finish the profile on the ID.

1. Select **Left ID**.



2. Click **Roughing**.
3. Click **System Default All**.
4. On the **General** tab:
   - Operation Name = **ID Rough and Finish**
   - Tool = **Rough 2H**
   - Speed CSS = **360**
   - Feedrate PR = **0.1**
5. On the **Strategy** tab:
   - Type of Work = **ID**
   - Finish Pass = **Yes**
   - Entry Mode = **Z Only**
   - Entry Point Z = **5.0**
   - Exit Mode = **Z Only**
   - Exit Point Z = **5.0**
6. On the **Rough** tab:
   - Stock Type = **Automation**
   - Rough Stock Z, X = **0.2**, **0.2**
   - Depth Variation = **Even Steps**
   - Maximum Depth of Cut = **2**
   - Depth Clearance = **5**
   - Lead-In Type = **None**
   - Lead-Out Type = **None**
7. On the **Finish** tab:
   - Finish CSS = **360**
   - Finish PR = **0.1**
   - Tangent Distance (Lead-In) = **2**
   - Tangent Distance (Lead-Out) = **2**

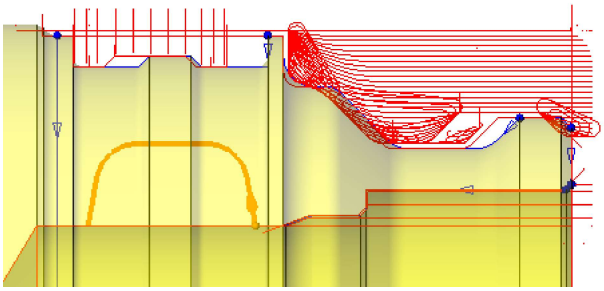
8. Click **OK**.



Groove the ID

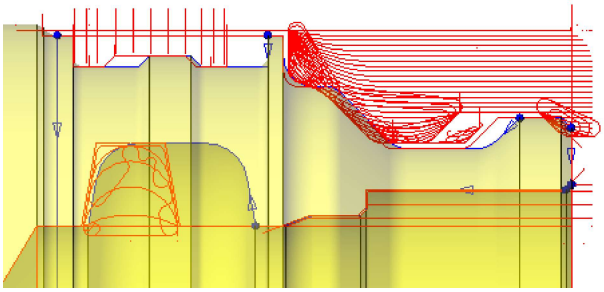
This grooving operation uses a round insert to cut the groove on the ID with a ProfitTurning strategy. Instead of plunge moves, the toolpath is rounded. The approach/exit moves have been changed to cut on the ID.

1. Select **Left Groove I.D.**.



2. Click **Grooving**.
3. On the **General** tab:
   - Operation Name = **ID Groove**
   - Tool = **ID Groove R4**
4. On the **Strategy** tab:
   - Type of Work = **ID**
   - Entry Mode = **Z Then X**
   - Entry Point Z, X = **5**, **9**
   - Exit Mode = **X Then Z**
   - Exit Point Z, X = **5**, **9**
   - Entry Exit Clearance = **1.0**
5. On the **Rough** tab:
   - Groove Type = **ProfitTurning**
6. On the **Finish** tab:
   - Lead-In Offset Z, X = **0**, **0**
   - Lead-Out Offset Z, X = **0**, **0**

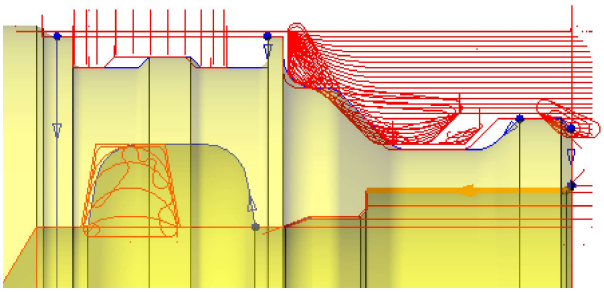
7. Click **OK**.



Thread the ID

Threading operations can be based on thread information that is entered manually or thread data can be loaded from an integrated threading database. In this operation, the threads will be created along the selected profile from manually entered data.

1. Select **Left I.D. Thread**.

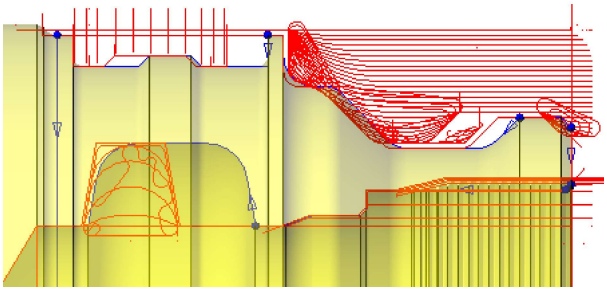


2. Click **Threading**.



3. Click **System Default All**.
4. On the **General** tab:
   - Operation Name = **ID Thread**
   - Tool = **ID Thread**
   - Speed CSS = **360**
   - Feedrate PR = **0.1**
5. On the **Strategy** tab:
   - Entry Mode = **Z Only**
   - Entry Point Z = **5.0**
   - Exit Mode = **Z Only**
   - Exit Point Z = **5.0**
   - Clearance = **.025**
   - Lead-Out Type = **Chamfer** (the lead-out will be angled)

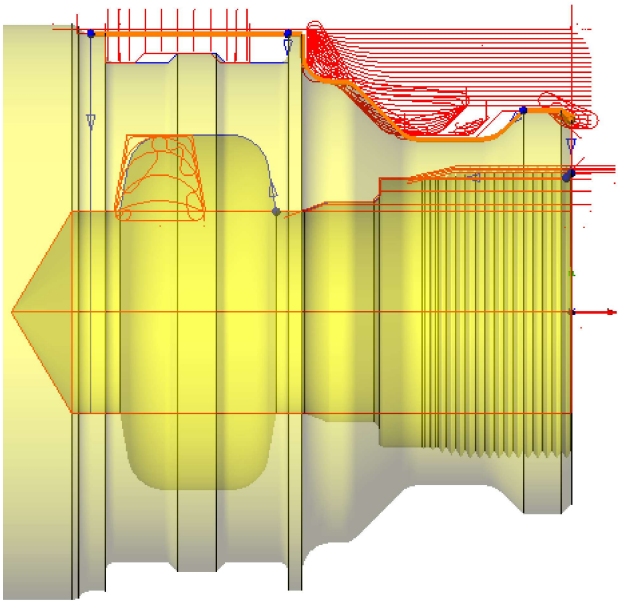
6. On the **Thread** tab:
   - Thread Depth = **1.2**
   - Major Diameter = **34.4** (press the Tab key to automatically calculate the minor diameter)
   - Threads/Lead = **2**
   - Start Length = **6** (the feature will be extended by 6 mm)
   - End Length = **-4** (a negative value shortens the feature)
   - First Depth of Cut = **0.6** (press the Tab key to automatically calculate the number of rough passes)
   - Multithread Cutting Mode = **Per Thread Offset Start Angle** (The two threads are each cut to full depth, but the second thread is offset by 180 degrees)
7. Click **OK.**



Perform a cutoff

The last step in this lesson is to separate the part from the bar stock.

1. Select the feature **Left O.D.**. The Cutoff operation will find the extremity point on the feature with the smallest Z value and use it as the start point for the operation



2. Click **Cutoff.**



3. Click **System Default All.**
4. On the **General** tab:

- Operation Name = **Cutoff**
- Tool = **Cutoff Tool** (notice that you can only choose tools that have two cutting edges)
- Speed RPM = **1910**
- Feedrate PM = **191**

5. On the **Rough** tab:
   - Rough to Diameter = **-1** (the tool will cut slightly below the center line)
6. Click **OK.**

